MARKUP LANGUAGES

SGML · HTML · XML

WHERE THIS FITS

Structure, Written Inside the Document Itself

Metadata	**Markup**
describes the object from the outside	**describes structure from the inside**

- Metadata sat beside the document — a separate record describing it (title, author, rights).
- **Markup is different: the structural information is embedded directly inside the text itself, as tags.**
- This is also the concrete machinery behind two things already taught: Structured Text Retrieval (chapters, sections, nodes) and Structural Queries (intitle:, site:) — markup is how that structure actually gets written down.

What Is a Markup Language?

```
<title>Information Retrieval Systems</title>
```

- **A markup language embeds tags directly inside plain text to describe its structure, meaning, or presentation — without changing the underlying content.**
- The word 'markup' comes literally from what a copy-editor once did on paper: annotate a manuscript with instructions ('make this bold', 'this is a heading') without touching the actual words.
- A tag names a role: <title> says 'this text is a title', not what the title should look like.

Why This Matters for Retrieval

- Markup is what turns a flat wall of text into an addressable structure — exactly the raw material Structured Text Retrieval models (Non-Overlapping Lists, Proximal Nodes) operate on.
- It is also what makes structural queries possible at all: a query like intitle:python can only work because the word 'python' is sitting inside an actual <title> tag somewhere.
- *Without markup, every document is just an undifferentiated string of characters — structure has to come from somewhere, and markup is where it's written down.*

SGML

- **Standard Generalized Markup Language — an ISO standard from 1986.**
- SGML is not itself a markup language you write documents in directly — it is a meta-language: a set of rules for defining other markup languages.
- A SGML-based language is defined by a DTD (Document Type Definition) — a formal rulebook stating exactly which tags exist and how they may be nested.
- **The single most important fact to remember about SGML:**
  - HTML itself was originally defined as an application of SGML — HTML is a language written using SGML's rules.

Real Life: SGML

DTD (rulebook)

<title> allowed once
<chapter> may repeat
<figure> needs a label



A Valid Document

must follow every
rule in the DTD

Still in active use today: aviation and defence manuals (ATA / S1000D standards) and long-lifespan government archives, where documents must outlive any one piece of software.

HTML

- **HyperText Markup Language — an SGML application built specifically for one job: describing web pages.**
- HTML has a fixed, pre-agreed vocabulary of tags — <title>, <p>, <img>, <a> — and nothing else. You cannot invent a new tag and expect a browser to understand it.
- It mixes structure and presentation quite freely — <b> literally means 'make this bold', a presentation instruction, sitting right next to <title>, a structural one.
- *This mixing is exactly the limitation that XML was built to fix — keep that tension in mind for two slides from now.*

Real Life: HTML

```
<html>
  <head><title>
IR Systems</title></head>
  <body>
    <h1>
Welcome</h1>
  </body>
</html>
```

HTML's Limitation

```
<b>Important</b>
```

presentation instruction
('make this bold')

```
<price>450</price>
```

structural / semantic tag
('this number is a price')

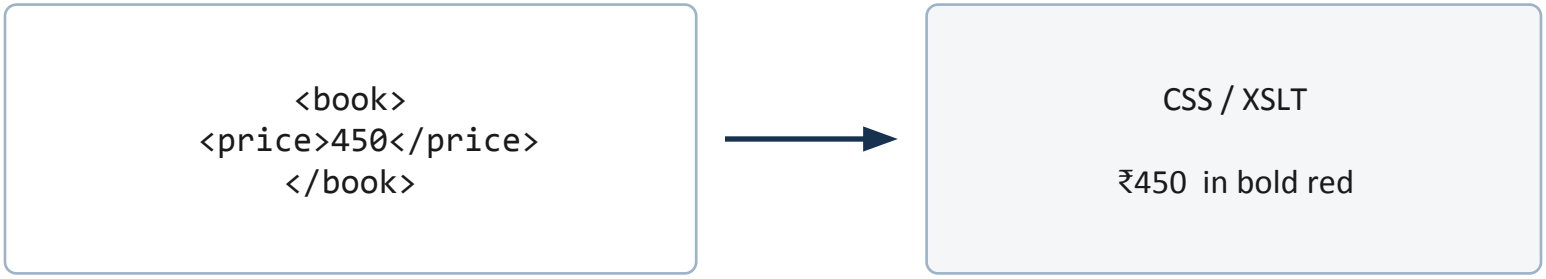
- HTML has no tag for <price> — you can only fake it with a fixed tag like and hope a human reads it correctly.
- **The fixed vocabulary is exactly the problem: HTML can say how something should look, but it has no way to say what something means.**

THE FIX

XML

- **eXtensible Markup Language — also an SGML-family language, but built to solve HTML's one big weakness.**
- There is no fixed tag vocabulary at all: you invent your own tags to name whatever your data actually is — <price>, <author>, <isbn>, <studentID> — anything.
- In exchange for that freedom, XML enforces strict, simple well-formedness rules: every tag opened must be closed, and tags must nest properly with no overlap.
- **The core design principle to give your class verbatim:**
  - XML separates structure/content from presentation — what something is, versus how it should look, are handled by two completely different tools.

Structure and Presentation, Kept Apart



XML — what it is

styling — how it looks

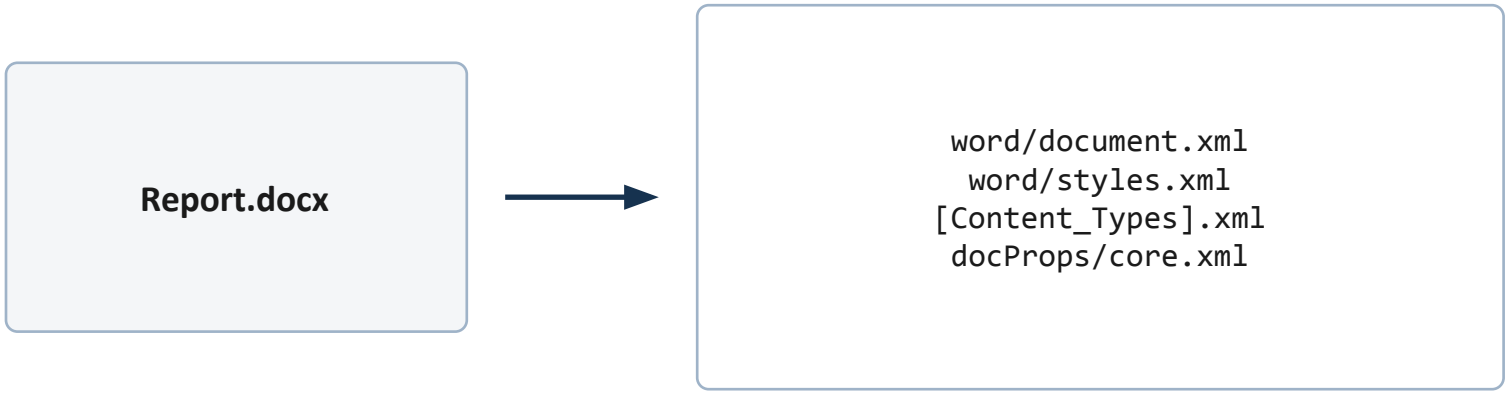
- The exact same XML file can be displayed completely differently on a phone, a printer, or an accessibility screen-reader — because the file itself carries no display instructions at all.
- That single XML file can also be read by a completely different program (a database, an accounting system) that never displays it at all — it only ever reads the structured data.

Real Life: RSS Feeds

```
<item>
  <title>
New IR Model Published</title>
  <pubDate>
10 Sept 2026</pubDate>
  <link>
https://...</link>
</item>
```

Every podcast app and news aggregator you've used pulls updates from exactly this — an XML file, refreshed automatically.

Real Life: Your Own Word Documents

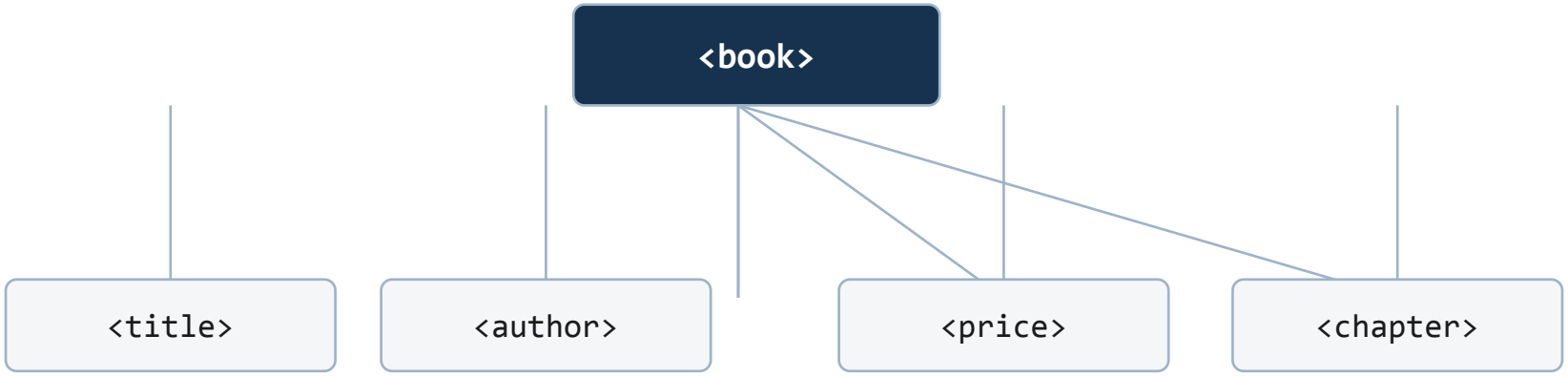


Every .docx, .xlsx, and .pptx file is literally a zip archive full of XML files — rename one to .zip and open it to see for yourself.

SGML vs. HTML vs. XML

	SGML	HTML	XML
Purpose	Meta-language for defining markup languages	Describing web pages for display	Describing and exchanging any structured data
Tag vocabulary	Defined per-DTD	Fixed, pre-agreed set	User-defined, unlimited
Structure vs. presentation	Not its concern	Freely mixed	Strictly separated
Still used for	Aerospace / defence manuals	Every web page	RSS, config files, Office file formats

The Thread Running Through Three Lectures



- **Markup**
  - writes this tree into the raw document.
- **Structured Text Retrieval**
  - (Non-Overlapping Lists, Proximal Nodes) searches within it.
- **Structural Queries**
  - (intitle:, site:) let a user ask for exactly one branch of it.

Putting It Together

Language	One-line identity	Real-life example
SGML	The meta-language that defines other markup languages	Aircraft manuals (S1000D)
HTML	Fixed vocabulary, mixes structure and presentation	Any web page's source
XML	User-defined tags, structure kept apart from display	RSS feeds, .docx files

Each one exists because the previous one had exactly one limitation the next was built to fix.